

Cross-Market LLM Disclosure Analysis: A Pilot Study of US and Indian Corporate Communications

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Abstract

Large language models are increasingly used to summarize, classify, and question-answer over financial disclosures. However, most benchmark intuition around financial NLP is shaped by US-style filings and earnings-call transcripts. This project investigates whether an LLM-based disclosure analysis pipeline performs similarly on US and Indian corporate communications. The system ingests earnings-call and concall transcripts, chunks and embeds the text, extracts sentiment, risk flags, topics, and financial signals, and presents results through a dashboard and retrieval-grounded chat interface. The pilot dataset contains 32 disclosure records across 13 sector-matched US/India company pairs, producing 128 benchmark chunks. Sentiment extraction is evaluated with per-class precision, recall, F1, macro-F1, and a US–India generalization gap.

The current MVP result shows a US macro-F1 of 0.717 and an India macro-F1 of 0.653, a gap of 0.063. This gap narrowed materially after adding fuller Indian source variants, suggesting that source fidelity is a major confound in cross-market financial NLP evaluation. Retrieval quality remains strong in both markets, with Recall@10 of 1.000 for US chunks and 0.984 for India chunks. The study is best understood as an educational pilot rather than a definitive benchmark because gold labels are AI-adjudicated and the sample remains small. Even so, the project demonstrates a working end-to-end research and product pipeline for cross-market disclosure intelligence.

1. Introduction

Financial disclosures are a major information source for investors, analysts, regulators, and researchers. Public companies explain performance, risks, guidance, and strategic priorities through earnings calls, concall transcripts, investor presentations, annual reports, and regulatory filings. These documents are long, semi-structured, and often written in market-specific language. As a result, they are natural candidates for retrieval-augmented generation and LLM-based information extraction.

However, a central question remains: does the same LLM pipeline work equally well across markets? Financial NLP systems are often evaluated on US-heavy data, where disclosure formats, terminology, and transcript conventions are relatively familiar. Indian corporate communications differ in format, unit conventions, regulatory vocabulary, and transcript availability. For example, Indian concall transcripts often mix qualitative management commentary with INR crore/lakh figures, sector-specific metrics, and analyst questions that differ from US earnings-call norms.

This project studies that issue through a practical MVP. It builds a full-stack disclosure analysis platform and uses it to compare extraction quality across US and Indian company disclosures. The goal is not only to create a useful dashboard, but also to measure whether an LLM-based pipeline generalizes across markets and where observed performance gaps may come from.

The main research question is:

Does an LLM-based sentiment and risk extraction pipeline perform equally well on US and Indian corporate disclosure text?

The current answer is nuanced. The pipeline performs reasonably well in both markets, but the measured India score initially lagged the US score. After adding fuller Indian source variants, the India macro-F1 improved and the cross-market gap narrowed. This suggests that source quality and transcript completeness are major confounds when comparing LLM performance across markets.

2. System Overview

The project is implemented as a full-stack retrieval-augmented analysis system. It has four main layers:

1. **Data pipeline:** raw disclosure text is collected, cleaned, chunked, embedded, and stored. 2. **LLM extraction layer:** sentiment, risks, topics, and financial signals are extracted from each chunk or document. 3. **Evaluation layer:** extracted sentiment is compared against benchmark labels, and retrieval quality is evaluated using a self-retrieval proxy. 4. **Application layer:** a FastAPI backend and Next.js frontend expose the results through a dashboard, benchmark page, company comparison page, risk explorer, and chat interface.

The data pipeline stores documents, chunks, vector embeddings, extraction results, benchmark labels, and financial snapshots in Supabase Postgres with pgvector. The backend exposes this data through typed API routes. The frontend presents the results in a lightweight research dashboard.

The MVP intentionally prioritizes transparency. The app distinguishes fuller transcript-derived records from summary-derived records through source-fidelity metadata, and the report discloses that labels are AI-adjudicated rather than independently human-labeled.

3. Dataset

The pilot dataset contains 32 disclosure records from 13 sector-matched US/India company pairs. The US side includes companies such as Microsoft, Apple, JPMorgan, Walmart, ExxonMobil, NVIDIA, Caterpillar, and Coca-Cola. The India side includes Infosys, TCS, HDFC Bank, Reliance Industries, Sun Pharma, ONGC, Tata Motors, Bharti Airtel, Hindustan Unilever, Bajaj Finance, Dixon Technologies, Larsen & Toubro, and Varun Beverages.

The corpus contains 128 benchmark chunks: 64 US chunks and 64 India chunks. Six India companies also have fuller source variants: Infosys, TCS, HDFC Bank, Reliance Industries, Larsen & Toubro, and Varun Beverages. These fuller records were added after the initial benchmark showed a larger US–India performance gap.

The dataset is deliberately small because this is a course-scale MVP. It is not intended to be a statistically definitive benchmark. Instead, it is designed to demonstrate the end-to-end workflow and surface practical issues that arise when applying LLM systems across markets.

4. Methodology

4.1 Chunking and Embedding

Raw disclosure text is chunked into approximately 150–220 word segments. Smaller chunks were used because some source records are summary-derived and would otherwise produce too few retrieval units. Each chunk is embedded using Gemini `gemini-embedding-001`, with output truncated to 768 dimensions. Embeddings and chunk metadata are stored in Supabase Postgres.

The project deliberately avoids approximate nearest-neighbor indexing at this scale. An earlier ivfflat index caused some relevant queries to return zero rows because the table was too small for the chosen index configuration. The current MVP uses exact similarity search through a sequential scan, which is simple and fast for 128 chunks.

4.2 Extraction

The extraction pipeline uses Groq-hosted `openai/gpt-oss-120b` generation for:

- sentiment classification,
- risk-flag extraction,
- topic extraction,
- financial-snapshot extraction,
- retrieval-grounded chat answers,
- AI-adjudicated benchmark labels.

For financial snapshots, the prompt explicitly instructs the model to report only figures stated in the collected text. If revenue, margin, EPS growth, free cash flow, or another metric is not directly stated, the field remains blank. This avoids fabricated financial values.

4.3 Benchmarking

Sentiment classification is evaluated using precision, recall, and F1 for positive, negative, and neutral classes. Macro-F1 is computed separately for US and India chunks. The cross-market generalization gap is defined as:

$$\text{Macro-F1(US)} - \text{Macro-F1(India)}$$

Retrieval quality is evaluated using a self-retrieval proxy. Each chunk's generated summary is used as a synthetic query, and the system checks whether the source chunk is retrieved in the top results. Metrics include Recall@1, Recall@3, Recall@5, Recall@10, mean reciprocal rank, and latency.

5. Results

5.1 Sentiment Classification

The current benchmark contains 128 labeled chunks. US chunks achieved a macro-F1 of 0.717 and India chunks achieved a macro-F1 of 0.653. The resulting cross-market gap is 0.063.

Market	n	Positive F1	Negative F1	Neutral F1	Macro-F1	Accuracy
US	64	0.935	0.600	0.615	0.717	0.844
India	64	0.778	0.400	0.783	0.653	0.766

The negative class remains the weakest class in both markets. This is expected because negative examples are rare in the dataset: the US subset has 7 negative examples and the India subset has 4. Small changes in negative-class predictions can therefore move macro-F1 substantially.

5.2 Retrieval Quality

Retrieval quality is strong overall, especially at higher k values.

Market	n	Recall@1	Recall@3	Recall@5	Recall@10	MRR	Mean latency
US	64	0.906	0.984	0.984	1.000	0.947	663 ms
India	64	0.734	0.922	0.969	0.984	0.827	655 ms

The lower India MRR likely reflects semantic overlap introduced by the fuller India variants. For example, summary and full-source records for the same company can contain similar themes, making the exact source chunk harder to retrieve at rank 1 even when a relevant chunk is retrieved.

6. Discussion

The most important project insight is that source quality strongly affects cross-market LLM evaluation. Earlier in the project, the India subset was more summary-derived, and the measured US–India macro-F1 gap was larger. After adding fuller Indian source variants, the India macro-F1 improved to 0.653 and the gap narrowed to 0.063.

This suggests that the initial performance difference should not be interpreted simply as "LLMs perform worse on Indian disclosures." A more careful interpretation is that models perform worse when source records contain less management tone, less Q&A context, fewer explicit figures, and less disclosure structure. Since source completeness differed by market, source fidelity became a confounding variable.

The result is useful because it changes what the next research step should be. Instead of immediately fine-tuning a model or changing prompts, the highest-value improvement is to build a more balanced full-source corpus for both markets. Only after source fidelity is balanced can the project more cleanly test whether remaining gaps come from terminology, market conventions, or model limitations.

7. Why the Project Is Unique and Useful

The project is unique because it combines an applied product with a benchmark question. Many student AI projects stop at a chat interface over documents. This project goes further by asking whether the pipeline is accurate, whether accuracy differs across markets, and what might explain that difference.

The system is useful in three ways:

1. **Disclosure intelligence:** users can browse sentiment, risk flags, topics, and financial snapshots across companies.
2. **Research benchmarking:** the app measures US vs India performance instead of assuming that one model works equally well everywhere.
3. **Source-fidelity analysis:** the project shows that incomplete source material can create an apparent model-performance gap.

This makes the MVP both a working dashboard and a methodological case study in responsible LLM evaluation.

8. Limitations

The project has several limitations. First, the benchmark labels are AI-adjudicated rather than independently human-labeled. This means the metrics should be treated as illustrative rather than as a rigorous empirical result. Second, the dataset is small: 128 chunks across 32 disclosure records. Third, source fidelity remains uneven even after adding six fuller India variants. Fourth, the financial snapshot extraction is intentionally sparse because the system only records numbers explicitly stated in the collected source text.

The project also uses a single model stack. Results may differ with other embedding models, generation models, prompt designs, or chunking strategies.

9. Future Work

Future work should focus on:

- adding full official source material for all India companies,
- adding more fiscal periods per company,
- replacing AI-adjudicated labels with independent human labels,
- adding a source-fidelity filter to compare summary-derived and full-transcript records,
- evaluating additional model providers,
- expanding retrieval evaluation with human-written queries and relevance judgments,
- improving financial extraction through official earnings releases and investor presentations.

The most important next step is full-source parity. Once both markets have comparable source quality, the benchmark can more fairly evaluate genuine cross-market model generalization.

10. Conclusion

This project demonstrates a working cross-market financial disclosure analysis platform and uses it to study LLM generalization across US and Indian corporate communications. The MVP ingests, embeds, extracts, evaluates, and serves disclosure intelligence through a dashboard and chat interface. The current result shows a small residual US–India sentiment gap, but the gap narrowed substantially after adding fuller Indian source records. The central conclusion is that source quality is a major confound in cross-market LLM evaluation. For practical financial NLP systems, better data collection may be just as important as better models.

References and Sources to Finalize

- Project benchmark outputs: `benchmark/metrics.json`, `benchmark/retrieval_metrics.json`.
- Project methodology: `docs/METHODOLOGY.md`.
- Project report: `docs/REPORT.md`.
- Gemini embedding model documentation.
- Groq/OpenAI-compatible generation model documentation.
- Supabase Postgres and pgvector documentation.
- Loughran-McDonald financial sentiment dictionary.
- Official company investor relations pages and transcript sources used in the corpus.